

The Annotated Rudin

A reader's companion to *Principles of Mathematical Analysis*

Third Edition

STYLE SAMPLE · OPTION A · B/W PRINT

Chapter 1, §§1.1–1.21 — roughly 8 of Rudin's pages

This sample exists so you can judge the *style* before I commit to eleven chapters. The mathematics is Rudin's; the boxes around it are the annotation layer. Everything about the layout — colours, box types, how much of Rudin's own text gets quoted, how deep the commentary goes — is still open.

How to read this book

Rudin is terse by design. He states things once, in their sharpest form, and leaves the reader to supply motivation, examples, and the missing algebra. This companion supplies exactly those three things and nothing else. It is meant to be read *beside* Rudin, not instead of him.

This book is set for black-and-white printing, so the boxes are told apart by their *frames*, not by colour. Five kinds appear throughout.

WHAT THIS CHAPTER IS FOR

A fully boxed panel with a solid black title bar opens every chapter. It says what the chapter is trying to achieve, gives the logical spine as a diagram, and marks which parts you can safely skip. A matching panel closes the chapter with the same ground shaded grey.

2.27 Theorem

A grey panel with a reverse-video number tab is Rudin, compressed. Definitions and theorem statements appear here in his own phrasing (lightly trimmed), so you always know what is his and what is mine. Proofs he gives are reproduced when the annotation needs to point at specific lines.

UNPACKING THIS

A **solid** rule down the left margin means unpacking. It answers “what is this actually saying,” “where did that formula come from,” and “why is the definition phrased *this* way and not the obvious way.” If you are reading Rudin for the first time, these are the boxes to read.

Going deeper

A **dashed** rule and a faint grey ground means the box goes further than the course requires: what breaks in a more general setting, which later theorem this quietly sets up, what the standard generalisation looks like. Skippable on a first pass.

PITFALL

Rules closing the box top and bottom, with a black label tab, mark a specific wrong belief that students reliably form here — and the counterexample that kills it.

READING THE PROOF

1. Longer proofs get a numbered walk-through like this one, keyed to Rudin’s lines. Each step says what is being done and, more importantly, *why that step and not another*.
2. Steps that look like rabbits pulled from hats — Rudin’s specialty —

get reverse-engineered, so you can see the choice being made rather than just verified.

Finally, margin notes like the one beside this paragraph carry short pointers that would interrupt the flow if inlined.

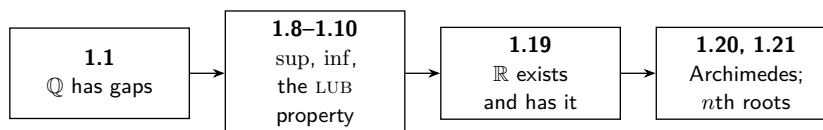
Margin notes carry one-line pointers: cross-references, the name of a result, a warning to slow down.

Chapter 1 The Real and Complex Number Systems

WHAT THIS CHAPTER IS FOR

Analysis is the study of limits, and a limit is a promise that some process *arrives* somewhere. The rationals break that promise: you can write down a sequence of rationals closing in on $\sqrt{2}$ with nothing at the end of it. Chapter 1 fixes this once, by isolating the single property $\mathbb{Q}$ lacks — the least-upper-bound property — and asserting a field that has it.

The logical spine is short:



Everything after 1.21 — the extended reals, $\mathbb{C}$, Euclidean space — is setting up notation for later chapters. The appendix builds $\mathbb{R}$ from $\mathbb{Q}$ by Dedekind cuts, discharging the promissory note in 1.19; you can read it whenever you like, or never, without affecting anything else.

1.1 The gap in $\mathbb{Q}$

Rudin opens by showing there is no rational p with $p^2 = 2$ — the familiar parity argument — and then does something less familiar. He sets

$$A = \{p \in \mathbb{Q} : p > 0, p^2 < 2\}, \quad B = \{p \in \mathbb{Q} : p > 0, p^2 > 2\},$$

and shows *A has no largest member and B has no smallest*. The tool is a formula that arrives with no explanation:

1.1 Example (the key construction)

To each rational $p > 0$ associate

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}, \quad (1.3)$$

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \quad (1.4)$$

If $p \in A$ then $p^2 - 2 < 0$, so (1.3) gives $q > p$ and (1.4) gives $q^2 < 2$; thus $q \in A$. If $p \in B$ then $p^2 - 2 > 0$, so $0 < q < p$ and $q^2 > 2$; thus $q \in B$.

WHERE DOES (1.3) COME FROM?

It is not magic and it is not Newton's method. Ask instead: *which* map $p \mapsto q$ would have $\sqrt{2}$ as a fixed point? Try the simplest shape, a ratio of linear terms $q = (ap + b)/(cp + d)$. Demanding $q = p$ at $p = \sqrt{2}$ means

$p^2 = 2$ must fall out of $p(cp + d) = ap + b$, and $q = (2p + 2)/(p + 2)$ is the smallest-integer solution: it gives $p^2 + 2p = 2p + 2$, i.e. $p^2 = 2$.

So (1.3) is the natural rational map pointing at $\sqrt{2}$. The reason it moves p toward $\sqrt{2}$ without ever crossing it is visible if you factor the error. A short computation gives

$$q - \sqrt{2} = \frac{\sqrt{2}(\sqrt{2} - 1)}{p + 2} (p - \sqrt{2}) = \frac{2 - \sqrt{2}}{p + 2} (p - \sqrt{2}).$$

For $p > 0$ the multiplier $(2 - \sqrt{2})/(p + 2)$ lies strictly between 0 and $(2 - \sqrt{2})/2 \approx 0.29$. Being *positive* is what keeps q on the same side of $\sqrt{2}$ as p — that is Rudin’s “ $q \in A$ ” and “ $q \in B$.” Being *less than one* is what makes the error shrink, which is why q is strictly better than p and no member of A can be largest.

This is a contraction, eight chapters early

The factorisation above says the map $p \mapsto (2p + 2)/(p + 2)$ contracts distance to $\sqrt{2}$ by a factor of at most 0.29 at every step. That is precisely the hypothesis of the *contraction principle*, 9.23, which Rudin uses to prove the inverse function theorem. Chapter 1 quietly runs the argument by hand before the general theorem exists.

Iterating from $p_0 = 1$ gives $1, \frac{4}{3}, \frac{10}{7}, \frac{24}{17}, \frac{58}{41}, \dots$ — the convergents of the continued fraction $\sqrt{2} = [1; 2, 2, 2, \dots]$, alternating below and above only because we started below. Compare Newton’s method on $f(p) = p^2 - 2$, which gives $p \mapsto (p^2 + 2)/(2p)$: faster (the error squares), but it always lands *above* $\sqrt{2}$, so it would prove “ B has no smallest” and say nothing about A . Rudin’s choice is the one that does both halves at once.

NO LARGEST MEMBER IS NOT OPENNESS

Openness is not the point and would not be enough. The content of 1.1 is that A is bounded above in $\mathbb{Q}$ — every element of B is an upper bound — yet among those upper bounds there is no *least* one, because B has no smallest element. A set can perfectly well have no largest member and still have a supremum: $\{r \in \mathbb{Q} : r < 0\}$ has none, but its supremum 0 exists in $\mathbb{Q}$. The failure in 1.1 is about the *upper bounds*, not about A .

1.8–1.10 Supremum, infimum, and the property that matters

1.8 Definition

Let S be an ordered set and $E \subset S$ bounded above. Suppose there is $\alpha \in S$ with

- (i) α is an upper bound of E ;
- (ii) if $\gamma < \alpha$ then γ is not an upper bound of E .

Then α is the *least upper bound*, or *supremum*, of E ; we write $\alpha = \sup E$.

Read 1.8 as two clauses: α is an upper bound, and nothing smaller is.

The infimum is defined symmetrically.

WHY NOT JUST SAY “THE LARGEST ELEMENT”?

Because it usually does not exist. $E_1 = \{r \in \mathbb{Q} : r < 0\}$ has no largest element, but the set of its upper bounds — everything ≥ 0 — does have a smallest, namely 0. The supremum relocates the question from E , where the answer is often “none,” to the set of upper bounds of E , where it is often “yes.”

Clause (ii) is the whole definition. It is stated in contrapositive form, and the version you will actually use in proofs is:

for every $\varepsilon > 0$ there is $x \in E$ with $x > \alpha - \varepsilon$.

Any $\gamma < \alpha$ fails to be an upper bound, which means *something in E sticks out above γ* . Reach for that sentence every time a proof begins “let $\alpha = \sup E$.”

1.10 Definition

An ordered set S has the *least-upper-bound property* if: whenever $E \subset S$ is nonempty and bounded above, $\sup E$ exists in S .

THE TWO HYPOTHESES ARE NOT DECORATION

“Nonempty” and “bounded above” are both load-bearing, and dropping either breaks the definition rather than merely weakening it. In $\mathbb{R}$, every real number is an upper bound of $\emptyset$, so $\emptyset$ has no least upper bound; and $\mathbb{Z} \subset \mathbb{R}$ is unbounded above, so it has none either. Any proof that produces a supremum owes you a check of both — in 1.21 below, Rudin spends two full sentences establishing exactly these before he writes $y = \sup E$.

1.11 Greatest lower bounds come for free

1.11 Theorem

Suppose S is an ordered set with the least-upper-bound property, $B \subset S$ is nonempty and bounded below, and L is the set of all lower bounds of B . Then $\alpha = \sup L$ exists in S , and $\alpha = \inf B$.

THE SHAPE OF THE TRICK

We are handed a hypothesis about suprema and asked for an infimum, so the only possible move is to find a set whose *supremum* is the infimum we want. The set of lower bounds L is the natural candidate: it sits entirely below B , so B pushes down on it from above, and its top edge is exactly

where B begins. The picture is worth more than the algebra:

$$\underbrace{\cdots \cdots \cdots}_L \bullet \underbrace{\cdots \cdots \cdots}_B \quad \text{with } \bullet = \sup L = \inf B.$$

The two sets meet at a single point, and the theorem says that point is both.

Rudin's proof. Since B is bounded below, L is not empty. Since L consists of exactly those $y \in S$ satisfying $y \leq x$ for every $x \in B$, every $x \in B$ is an upper bound of L . Thus L is bounded above, and the hypothesis on S gives $\alpha = \sup L$ in S .

If $\gamma < \alpha$ then γ is not an upper bound of L , hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$; thus $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L . So $\alpha \in L$ but $\beta \notin L$ whenever $\beta > \alpha$; that is, α is a lower bound of B and no larger element is. Hence $\alpha = \inf B$. $\square$

READING THE PROOF

1. $L \neq \emptyset$, L **bounded above**. Both hypotheses of the LUB property must be checked before invoking it. Nonempty is the assumption that B is bounded below, restated. Bounded above is the observation that every single $x \in B$ is an upper bound for L — by the very definition of “lower bound of B .” This is the one genuine idea in the proof.
2. $\alpha \in L$, i.e. α is a **lower bound of B** . Rudin argues contrapositively and compresses two steps into one line. Unpacked: take any $\gamma < \alpha$. Since α is the *least* upper bound of L , γ is not an upper bound of L , so some $y \in L$ has $y > \gamma$. But y is a lower bound of B , so $\gamma < y \leq x$ for all $x \in B$, and in particular $\gamma \notin B$. No element of B is therefore less than α , which is to say $\alpha \leq x$ for all $x \in B$.
3. **Nothing bigger works**. If $\beta > \alpha$ then $\beta \notin L$, because α is an upper bound of L . So β is not a lower bound of B . Combined with step 2, α is a lower bound and no larger element is — which is the definition of $\inf B$.

Why this is not a triviality

The mirror-image statement is false in general order-theoretic settings, and the reason the proof works here is that L and B are *complementary* in a strong sense: the upper bounds of L include all of B . Take the greatest lower bound property as a hypothesis instead and the same argument runs backwards, so the two properties are equivalent for ordered sets. This is why Rudin never states a separate “greatest-lower-bound property” axiom for $\mathbb{R}$: 1.19 gives him one, and 1.11 hands him the other. Every later use of $\inf$ — lower Riemann–Stieltjes sums in 6.2, $\liminf$ in 3.16 — rests on

■ this one theorem.

1.21 n th roots, and the rabbit out of the hat

This is the theorem that repays the whole chapter: it is the first time the LUB property produces a number that $\mathbb{Q}$ could not.

1.21 Theorem

For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y with $y^n = x$.

Rudin sets $E = \{t > 0 : t^n < x\}$, checks E is nonempty and bounded above, puts $y = \sup E$, and rules out $y^n < x$ and $y^n > x$ in turn. The first case runs like this:

1.21 Proof, the case $y^n < x$

The identity $b^n - a^n = (b - a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$ yields

$$b^n - a^n < (b - a)n b^{n-1} \quad (0 < a < b). \quad (*)$$

Assume $y^n < x$. Choose h so that $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y + 1)^{n-1}}.$$

Put $a = y$, $b = y + h$. Then

$$(y + h)^n - y^n < h n (y + h)^{n-1} < h n (y + 1)^{n-1} < x - y^n,$$

so $(y + h)^n < x$ and $y + h \in E$. Since $y + h > y$, this contradicts the fact that y is an upper bound of E .

REVERSE-ENGINEERING THE CHOICE OF h

Nobody guesses that h . It is found by writing down the goal and solving backwards; Rudin simply deletes the derivation. Here it is.

Goal. Find $h > 0$ with $(y + h)^n < x$. Equivalently, $(y + h)^n - y^n < x - y^n$, where the right-hand side is a fixed positive number because we assumed $y^n < x$.

Bound the left side. We cannot compute $(y + h)^n - y^n$, but $(*)$ with $a = y$, $b = y + h$ bounds it: $(y + h)^n - y^n < h n (y + h)^{n-1}$. So it is *enough* to make $h n (y + h)^{n-1} \leq x - y^n$.

Remove h from the exponent. That inequality still has h buried inside $(y + h)^{n-1}$, which we cannot solve for. Kill it by agreeing in advance that $h < 1$; then $(y + h)^{n-1} < (y + 1)^{n-1}$, a constant. This is the only reason the condition “ $0 < h < 1$ ” appears — it is bookkeeping, not mathematics.

Solve. It now suffices that $h n (y + 1)^{n-1} < x - y^n$, i.e.

$$h < \frac{x - y^n}{n (y + 1)^{n-1}},$$

which is Rudin's line. Both conditions on h are upper bounds, so take h below the smaller of the two and such an h exists.

What the proof is really doing

Strip the algebra and the argument is: $t \mapsto t^n$ is continuous, so if y^n were strictly below x we could nudge y upward and stay below x . The displayed choice of h is a hand-rolled continuity estimate, written out because continuity is not available until 4.4 — and 4.4's machinery is not available until $\mathbb{R}$ exists, which is what we are in the middle of establishing. The same argument reappears in clean form as an application of the intermediate value theorem (4.23) once Chapter 4 is in place. Note also which side each case attacks. Assuming $y^n < x$ contradicts " y is an *upper bound*"; assuming $y^n > x$ contradicts " y is the *least upper bound*." Both halves of 1.8 get used exactly once, which is a good sign that the choice of E was the right one.

WHY E NEEDS $T > 0$

Without the positivity restriction, $E = \{t : t^n < x\}$ is unbounded below for odd n and the set is a different animal for even n . More to the point, the uniqueness claim is false without it: for $n = 2$, $x = 4$, both 2 and -2 are roots. Rudin's "one and only one *positive*" is doing real work.

CHAPTER 1 IN ONE PARAGRAPH

$\mathbb{Q}$ fails to be a home for analysis because bounded sets can have no least upper bound (1.1). Name that failure (1.8–1.10), assert a field free of it (1.19), and the consequences follow immediately: infima exist too (1.11), $\mathbb{N}$ is unbounded and $\mathbb{Q}$ is dense (1.20), and every positive real has every root (1.21). Nothing in the rest of the book uses more about $\mathbb{R}$ than this. When a later proof says "by completeness," it means 1.19 and nothing else.